

51143339 FU-LISYUW

(1)

1. Counter = CD (B=Jump, R₀=R₀ ⇒ ✓)

2.

(a.) 7123

OR the contents of register 2 and 3 to be placed in register 1.

(b.) 40E1

register E to be copied into register 1.

(c.) A304

Rotate the bit pattern in register 3 four times to the right, in a circular fashion.

(d.) B100

Compare 1 with the contents of register 0. If the two were equal, the pattern 0 would be placed in the PC. Otherwise the program would continue.

(e.) 2B1D

(program counter)

LOAD the register B with the bit pattern CD.

3.

(a.) 2677 (b.) 3777 (c.) BA24 (d.) A403 (e.) 81E2

4.

(a.)

2655 $\Rightarrow$ LOAD the register 6 with the bit pattern 55

(b.)

101 101 (55)

5.

20(00) 04(01) : $R_0 = 04$

(a) register 0 : 04

21(02) 01(03) : $R_1 = 01$

(b) register 1 : 04

40(04) 12(05) : $R_1 \rightarrow R_2 = 01$

(c) counter = 06

51(06) 12(07) : $R_1 + R_2 = \text{new } R_1(02)$

B1(08) 0C(09) : $04 \neq 02$

B0(0A) 06(0B) : $04 = 04 \Rightarrow M[06]$

51(06) 12(07) : $R_1 + R_2 = \text{new } R_1(03) \Rightarrow R_1 = 04$
again

B1(08) 0C(09) : $04 = 04 \Rightarrow M[0C]$

60(0C) 00(0D) : halt

6.

12



12(20) 20(21) : $R_2 \leftarrow M[20] = 12(18)$

32(22) 30(23) : $M[30] \leftarrow R_2 = 12(18)$

B0(24) 21(25) : $R_0 = R_0 \Rightarrow PC: 21$

20(26) 32(27) : $R_0 = 32(50)$

30(28) B0(29) : $M[B0] \leftarrow R_0 = 32(50)$

21(30) 24(31) : $R_1 = 24(36)$

20(32) 00(33) : halt

(a) $R_0 = 32$, $R_1 = 24$, $R_2 = 12$

(b) $M[30] = 12$

(c) $M[B0] = 32$

7.

25(00) B0(01) : $R_5 = B_0$

35(02) 04(03) : $R_5 \rightarrow M[04](B_0)$

B0(04) 00(05) : $R_0 = R_0 \Rightarrow PC: 00$

25(00) B0(01) : $R_5 = B_0$

35(02) 04(03) : $R_5 \rightarrow M[04](B_0)$

B0(04) 00(05) : $R_0 = R_0 \Rightarrow PC: 00$

Loop (can't stop)

8. value (hexadecimal)

$$20_{(30)} \ 03_{(31)} : R_0 = 03_{(3)}$$

$$21_{(32)} \ 01_{(33)} : R_1 = 01_{(1)}$$

$$22_{(34)} \ 00_{(35)} : R_2 = 00_{(0)}$$

$$23_{(36)} \ 10_{(37)} : R_3 = 10_{(16)}$$

$$14_{(38)} \ 00_{(39)} : R_4 \leftarrow M[00] = 03_{(3)}$$

$$34_{(3A)} \ 10_{(3B)} : R_4 \rightarrow M[10] = 03_{(3)}$$

$$52_{(3C)} \ 21_{(3D)} : R_2 + R_1 = \text{new } R_2 = 01_{(1)}$$

$$53_{(3E)} \ 31_{(3F)} : R_2 + R_1 = \text{new } R_3 = 11_{(17)}$$

$$32_{(40)} \ 39_{(41)} : R_2 \rightarrow M[39] = 01_{(1)}$$

$$33_{(42)} \ 3B_{(43)} : R_3 \rightarrow M[3B] = 11_{(17)}$$

$$B2_{(44)} \ 48_{(45)} : 03 \neq 02$$

$$B0_{(46)} \ 38_{(47)} : 03 = 03 \Rightarrow M[38]$$

$$14_{(38)} \ 00_{(39)} : \dots$$

$$34_{(3A)} \ 10_{(3B)} : \dots$$

$$52_{(3C)} \ 21_{(3D)} : R_2 + R_1 = \text{new } R_2 = 02_{(2)} \xrightarrow{\text{again}} 03_{(3)}$$

$$53_{(3E)} \ 31_{(3F)} : R_2 + R_1 = \text{new } R_3 = 12_{(18)} \xrightarrow{\text{again}} 13_{(19)}$$

$$32_{(40)} \ 39_{(41)} : R_2 \rightarrow M[39] = 03_{(3)}$$

$$33_{(42)} \ 3B_{(43)} : R_3 \rightarrow M[3B] = 13_{(19)}$$

$$(a) M[10] = 03_{(hex)} = 3_{(dec)}$$

$$(b) M[39] = 03_{(hex)} = 3_{(dec)}$$

$$(c) M[3B] = 13_{(hex)} = 19_{(dec)}$$

9.

- (a) 01001 (b) 000000 (c) 000100
(d) 110011 (e) 111001 (f) 111110
(g) 010101 (h) 111111 (i) 010000
(j) 101101 (k) 000101 (l) 001010

10.

(a)

$$3F = 00111111 \rightarrow 11001111 = CF_{(207)}$$

(b)

$$DD = 00001101 \rightarrow 01000011 = 43_{(67)}$$

(c)

$$FF = 11111111 \rightarrow 11111111 = FF_{(255)}$$

(d)

$$77 = 01110111 \rightarrow 11011101 = DD_{(221)}$$